employment standards; continuous monitoring feeds ethical and legal oversight processes.

Figure 5.1 provides a visual representation of the core architecture of fourth-generation drones, illustrating how modern UAVs integrate sensing, autonomy, and hardware systems to achieve intelligent and adaptive operations. The diagram highlights three major functional domains: sensing and perception, autonomy, and hardware. In the sensing section, components such as electro-optical/infrared (EO/IR) cameras, onboard radar, and LiDAR support high-precision environmental awareness and object detection. The autonomy section depicts AI-based decision-making, automated object detection, and swarm coordination, enabling drones to collaborate, navigate, and execute missions with minimal human input. The hardware section emphasizes AI-driven flight control, beyond visual line of sight (BVLOS) communication, long-endurance airframes, and energy-dense batteries, which collectively enhance range, reliability, and mission persistence. Together, the figure encapsulates the essence of fourth-generation UAV design, a fusion of intelligent perception, adaptive autonomy, and advanced aerostructural engineering for multi-domain, next-generation aerial operations.